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Laboratory #6: Hierarchy Layout of an XOR gate		

I. OBJECTIVES

- To create a layout for the Two-input XOR gate
- To perform DRC and LVS.

II. LAB PROCEDURE

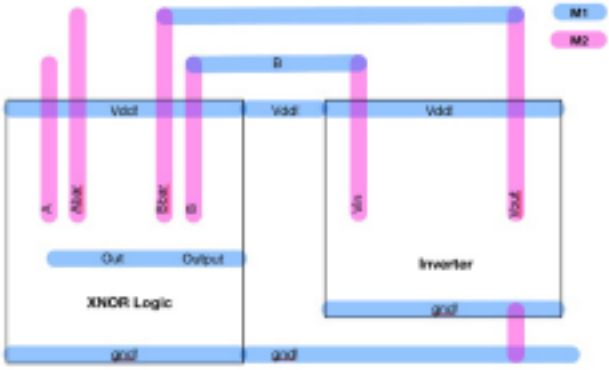
1. XOR Layout Lab Flow

As a part of Hierarchy design, this lab is divided into two parts.

- XOR Logic (This has four Inputs i.e., A, Abar, B, Bbar)
- Four Inputs to Two Inputs using Two Inverters.

2. XOR Logic Layout

- Open the Library that you created last week for the XOR Logic gate.
- Create a layout view for your XOR Logic gate. Remember that your layout view should be within the same XOR Logic cell that you created the schematic view in.
- Use the tutorial as a reference for different steps. Also, you may refer to the Design rules Excel sheet provided by your T.A.
- Create the Two-Input XOR floorplan and layout as shown below, using the XOR Logic cell, and Inverter cells.
- To add the XOR Logic and Inverter gate layout. Select Create→ instance, or press " i" for its keyboard shortcut. Then select the layout you desire and place it as shown below floor plan.



Floor Plan - XNOR Logic and Inverter

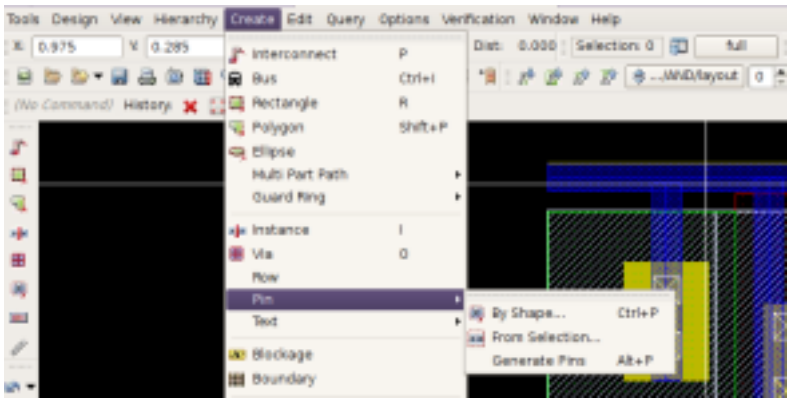
NOTE: Above floor plan is incomplete, and it is up to student to figure out the rest with the help of TA.

- Creating pins is different in hierarchy design.

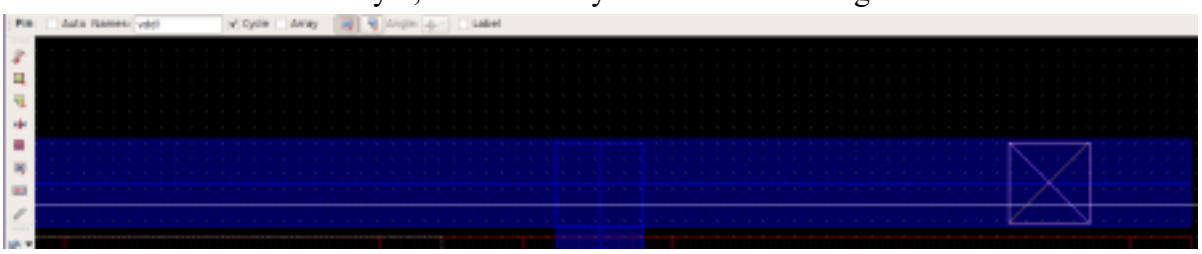
- **Adding Terminals/Pins**: In order to clearly show input/output terminals when used in the hierarchy to build the next stages, it is important to also add Terminals. Follow this procedure to create them:

  a) It involves two steps. Creating a **Pin** and placing a **Label**

  b) **Pin:** Click Create → Pin → By Shape.



  c) Add the name of your pin, in this example we are creating the vdd! Pin. And then use the same M1 or M2 layer to draw a metal rectangle. Since vdd! is made of M1 layer, so use M1 layer as a metal rectangle.



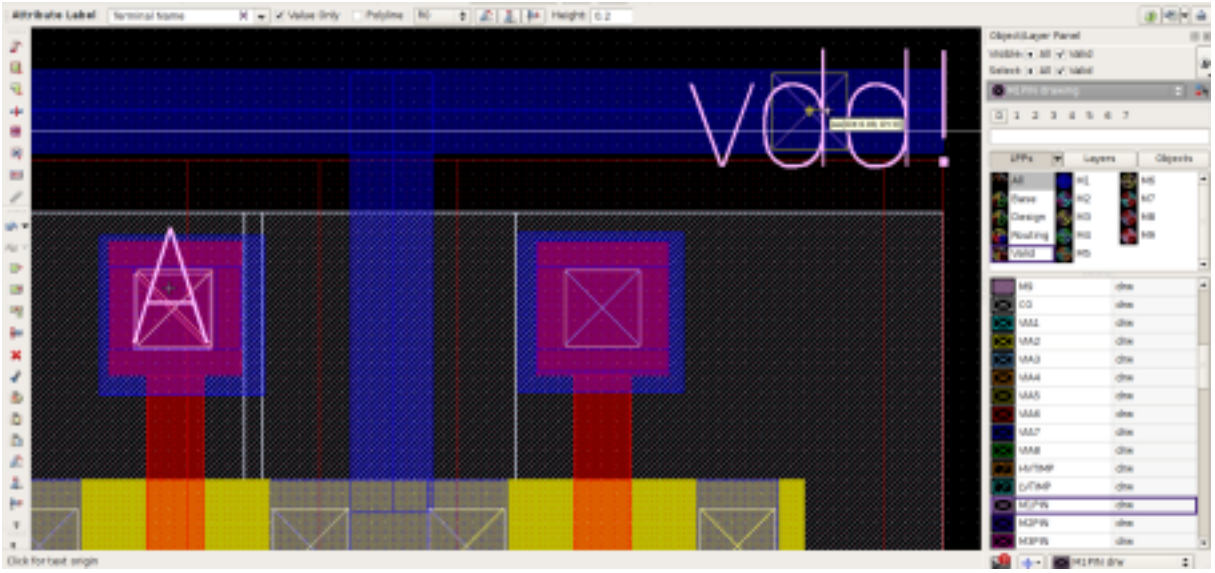
  d) **Label:** Select Create → text → attribute label (keyboard shortcut shift+l).

  e) From the drop down menu select Terminal Name, and select the metal rectangle you drew in step c.

  f) The vdd! Label will show up, but will be in metal layer by default, before placing it, adjust the label to the proper metal pin. (M1pin/M2pin). As this is for vdd! made if M1 layer then use M1 Pin layer as a label.

  g) Place the label.

  h) Repeat these steps for all your input/output pins.



3. DRC and LVS

- Perform a Design Rules Check (DRC) as well as a Layout vs. Schematic (LVS) test
- Adjust your layout if necessary to fix any errors.
- Refer to your tutorial as a reference for different steps. Also note that the tutorial includes a troubleshooting guide.

III. <u>REPORT</u>

Write a short laboratory report that details all the work done. Describe the objective and procedures of this lab with your own words. The lab report should contain the following:

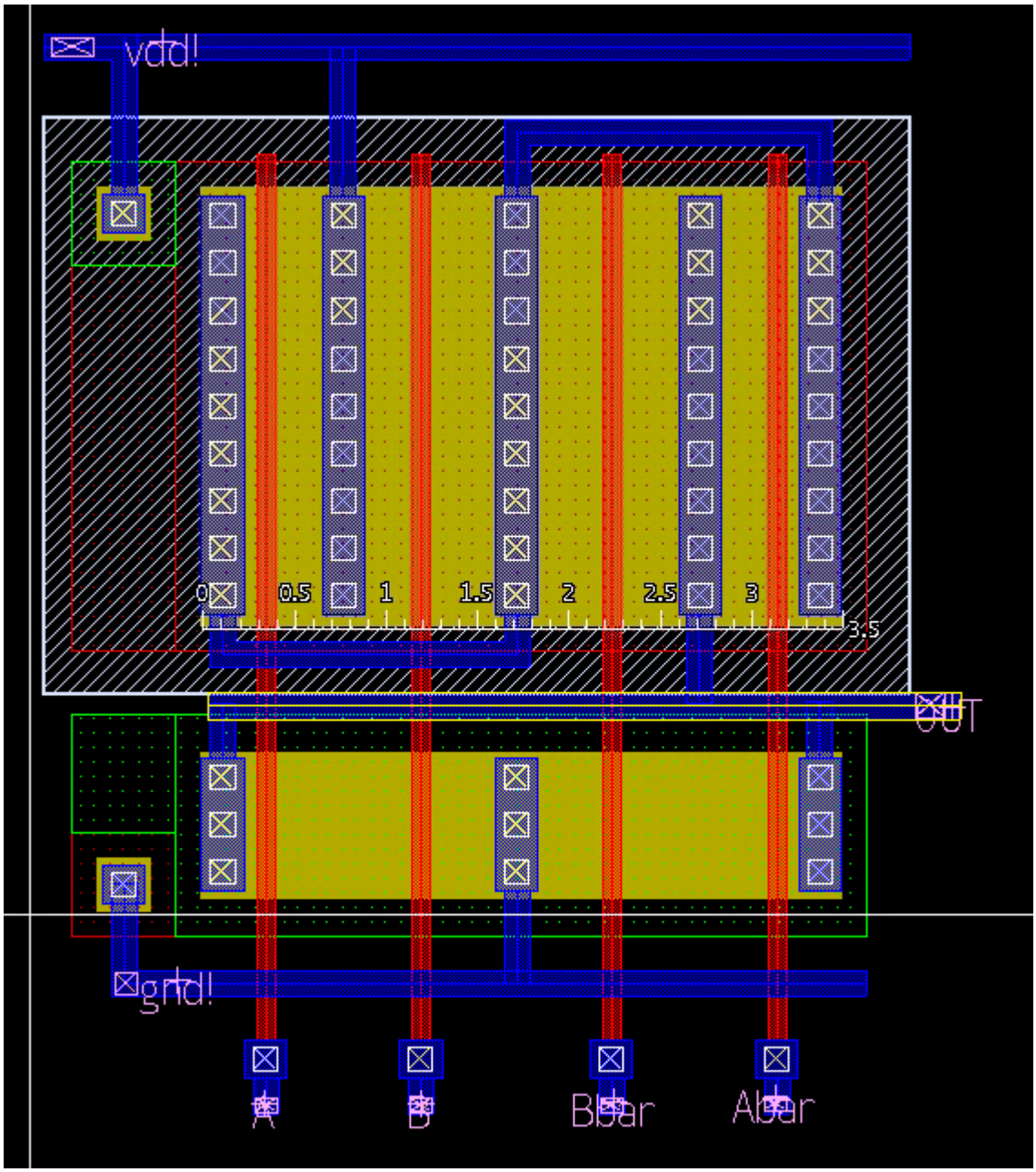
a) All schematics and layouts are used in your lab.

b) Screenshots showing your design passed DRC and LVS

c) Answer any questions in the lab assignment.

d) Conclusions

Part 1: 4 Input Xor


vdd!
0
0.5
1
1.5
2
2.5
3
3.5
OUT
gnd!
A
B
Bbar
Abar

File View Tools Windows Help

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LAYOUT ERRORS RESULTS

CLEAN

DRC Error Statistics

```
Library name:    Lab5
Structure name:  XorGate
Generated by:    IC Validator RHEL64 V-2023.12-SP3.10109947 2024/04/19
Runset name:     /usr/synopsys/SAED/SAED90nm_PDK/icv/drc/rules.drc.9m_saed90_icv.rs
User name:       dthornburg
Time started:    2025/05/15 06:29:43PM
Time ended:      2025/05/15 06:29:49PM

Called as: icv -f openaccess -i Lab5 -c XorGate -oa_view layout -oa_lib_defs /DCNFS/users/student/dthornburg/ELEN153L/lib.defs
```

LVS Compare Results: PASS

DRC and Extraction Results: CLEAN

[XorGate, XorGate]

Netlist Extraction Statistics

```
Library name:    Lab5
Structure name:  XorGate
Generated by:    IC Validator RHEL64 V-2023.12-SP3.10109947 2024/04/19
Runset name:     /usr/synopsys/SAED/SAED90nm_PDK/icv/lvs/rules.lvs.9m_saed90_lvs.rs
User name:       dthornburg
Time started:    2025/05/15 06:42:34PM
Time ended:      2025/05/15 06:42:49PM

Called as: icv -f openaccess -i Lab5 -c XorGate -oa_view layout -oa_lib_defs /DCNFS/users/student/dthornburg/ELEN153L/lib.defs
```

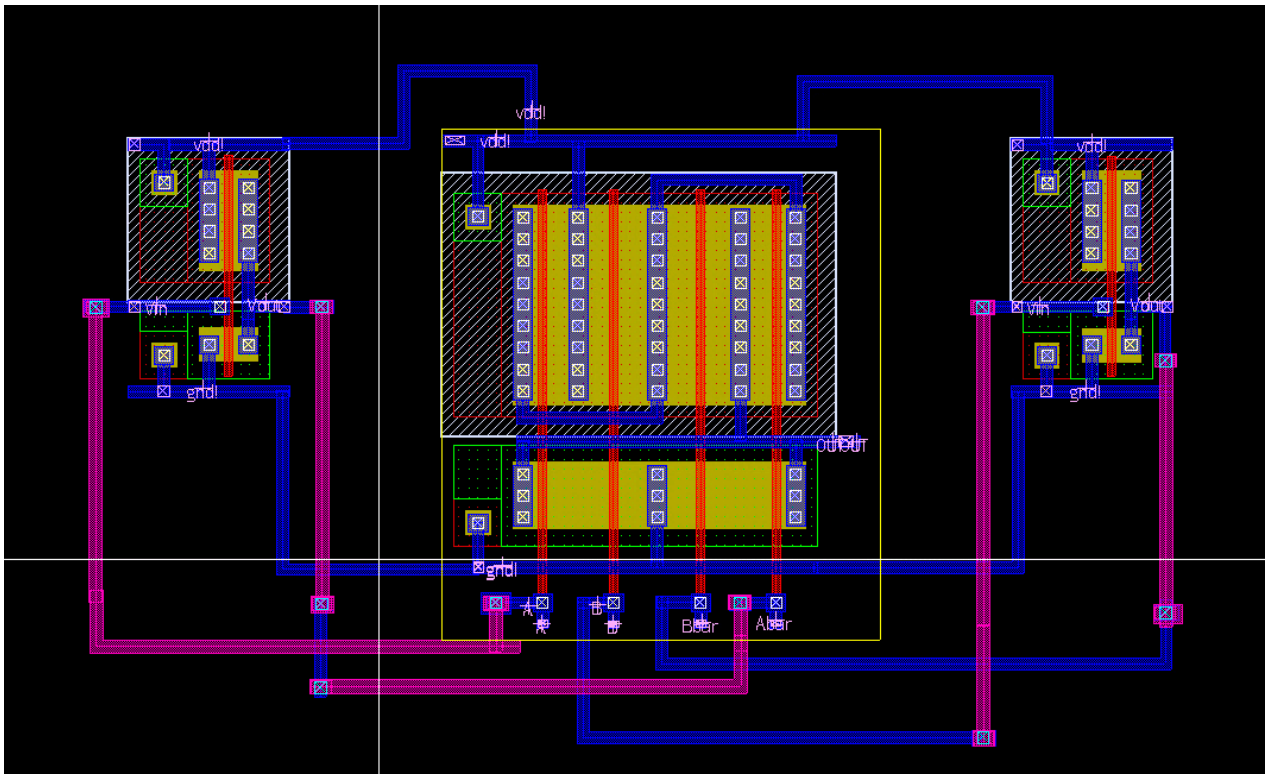
Layout vs. Schematic Statistics

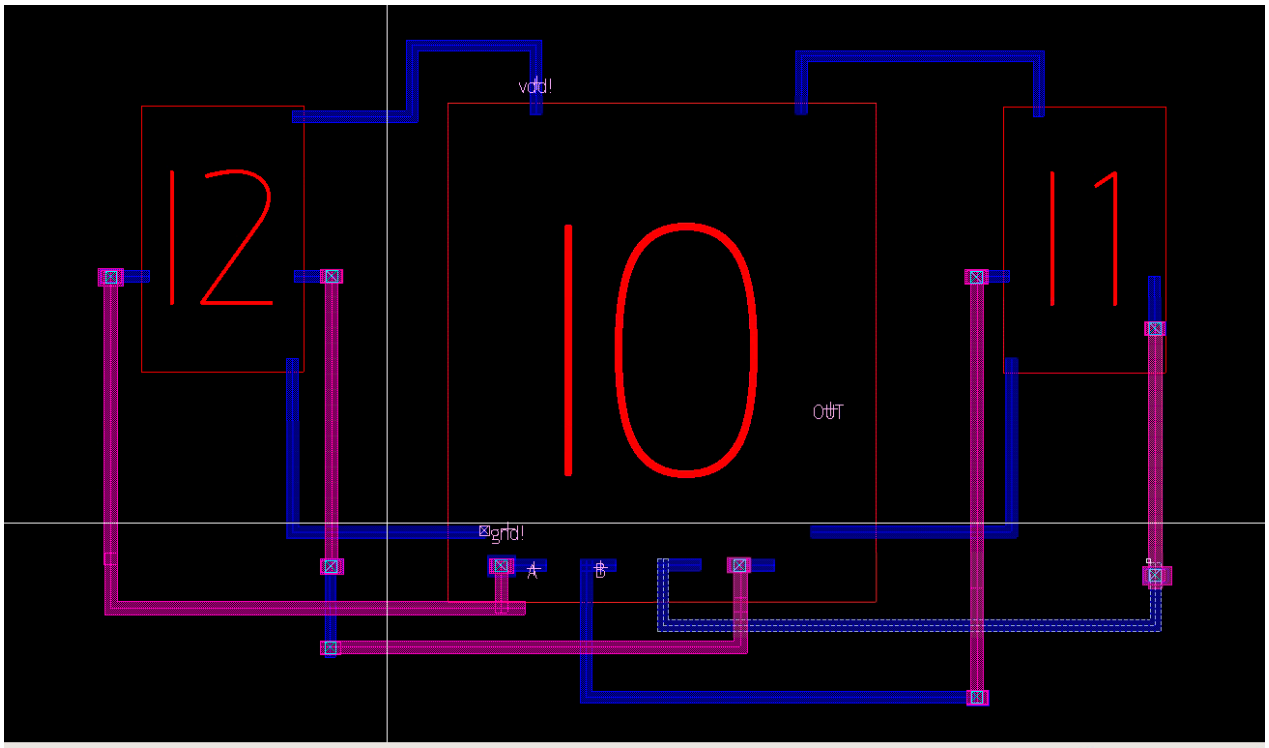
```
Schematic: /DCNFS/users/student/dthornburg/ELEN153L/synopsys_custom/pvjob_Lab5.XorGate.icv.lvs/XorGate.sch_out.gz
```

LVS Errors:

```
  blackbox cells checked: 0
    passed  0
    failed  0
  equivalence points checked: 1
    passed  1
    failed  0
```

Part 2: Two Input XOR


vdd!
vdd!
vdd!
vdd!
vin
Vout2
vin
Vout2
gnd!
gnd!
OUT
gnd!
A
B
Bbar
Abar


vdd!
12
10
11
OUT
gnd!
A
B

VUE: [/DCNFS/users/student/dthornburg/ELEN153L/synopsys_custom/pvjob_Lab5.XorGate2.icv.drc/XorGate2.vue]

File View Tools Windows Help

Load Results × | Run Summary ×

LAYOUT ERRORS RESULTS

CLEAN

DRC Error Statistics

```
Library name:    Lab5
Structure name:  XorGate2
Generated by:    IC Validator RHEL64 V-2023.12-SP3.10109947 2024/04/19
Runset name:     /usr/synopsys/SAED/SAED90nm_PDK/icv/drc/rules.drc.9m_saed90_icv.rs
User name:       dthornburg
Time started:    2025/05/15 07:20:36PM
Time ended:      2025/05/15 07:20:41PM

Called as: icv -f openaccess -i Lab5 -c XorGate2 -oa_view layout -oa_lib_defs /DCNFS/users/student/dthornburg/ELEN153L/lib.def
```

LVS Compare Results: PASS

DRC and Extraction Results: CLEAN

[XorGate2, XorGate2]

Netlist Extraction Statistics

```
Library name:    Lab5
Structure name:  XorGate2
Generated by:    IC Validator RHEL64 V-2023.12-SP3.10109947 2024/04/19
Runset name:     /usr/synopsys/SAED/SAED90nm_PDK/icv/lvs/rules.lvs.9m_saed90_lvs.rs
User name:       dthornburg
Time started:    2025/05/15 07:56:39PM
Time ended:      2025/05/15 07:57:05PM

Called as: icv -f openaccess -i Lab5 -c XorGate2 -oa_view layout -oa_lib_defs /DCNFS/users/student/dthornburg/ELEN
```

Layout vs. Schematic Statistics

```
Schematic: /DCNFS/users/student/dthornburg/ELEN153L/synopsys_custom/pvjob_Lab5.XorGate2.icv.lvs/XorGate2.sch_out.g
```

LVS Errors:

```
    blackbox cells checked: 0
      passed  0
      failed  0
    equivalence points checked: 1
      passed  1
      failed  0
```